pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Course 5C Capstone

WEEK 2 · DAY 5

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The brief

The desk's question, in the past tense

The risk models of the book (200 invented bonds) and the cohort (4,000 invented firms) were built on the development years: the curve to 30 Dec 2022, study years 1 to 14. When the held-out years were opened, 939 dates to 2 Oct 2026 and study years 15 to 20, which models held, and which failed?

The workflow

1. Credit loss: Monte Carlo, VaR, shortfall
2. Historical VaR and its backtest
3. A hazard model, chosen by a tested rule
4. A screen of planted trades
5. Every choice committed
6. The held-out years, opened once

A model that fails on the held-out years is a result, and you say so in plain words.

Names for a model's output

The words for what a model gives, in this capstone as on every day:

Allowed names	Never called
a simulated loss, VaR, a shortfall	a forecast of a loss
the model's default chance	a real issuer's default chance
a flag: a print to check	mispriced, or a view on a bond

A historical VaR reads: had these days' moves repeated on the book of 30 Sep 2026.

The order of the work



and 18,503 firm-years: fit, choose

tested, committed

and 11,112 firm-years, opened once

The notebook enforces it

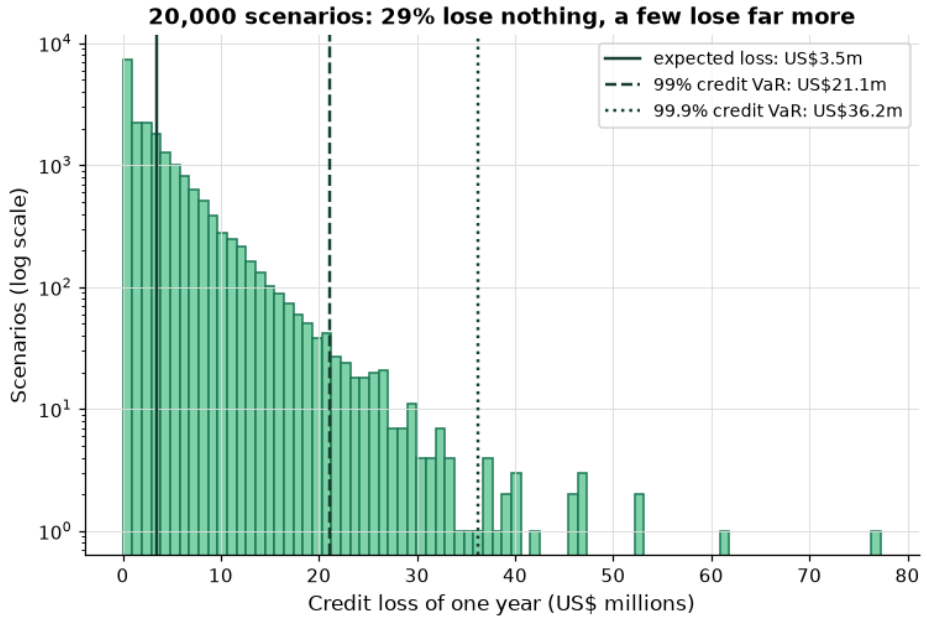
The loading cell reads the development rows only. open_held_out_rows() stops before its first use unless choice.md is committed, and stops on a second use.

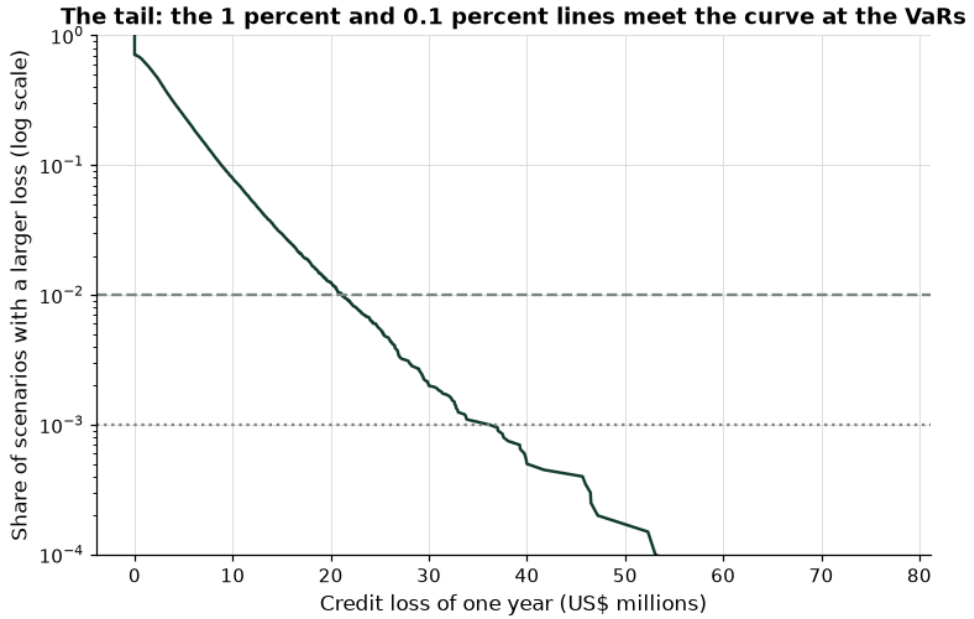
Why

Opening the years twice makes them more development years. If the choice cannot be shown to come before the score, the score is not a test of the choice.

The choice has a date on it, and the date is Git's.

Monte Carlo credit loss

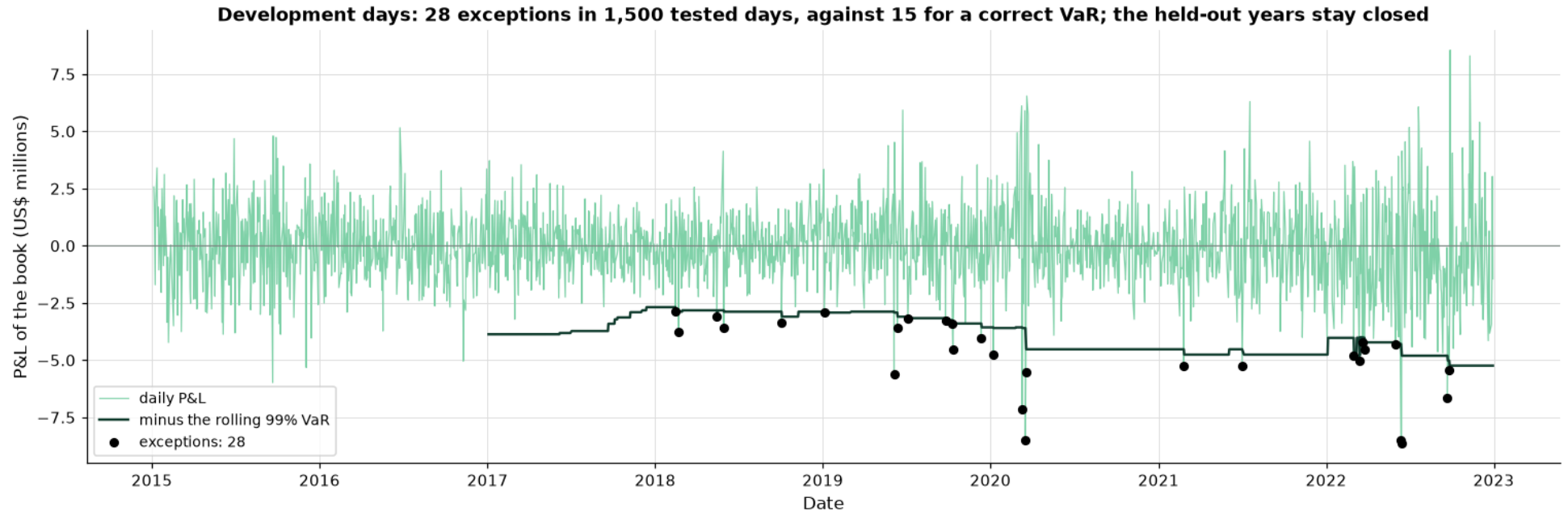




Level	Credit VaR	Expected shortfall
99%	US$21.1m	US$27.6m
99.9%	US$36.2m	US$45.2m

Expected loss US$3,482,412 a year; 29.2% of the scenarios lost nothing. Inputs chosen for the lesson: rho 0.2, LGD a beta with mean 0.6.

Historical VaR of the book



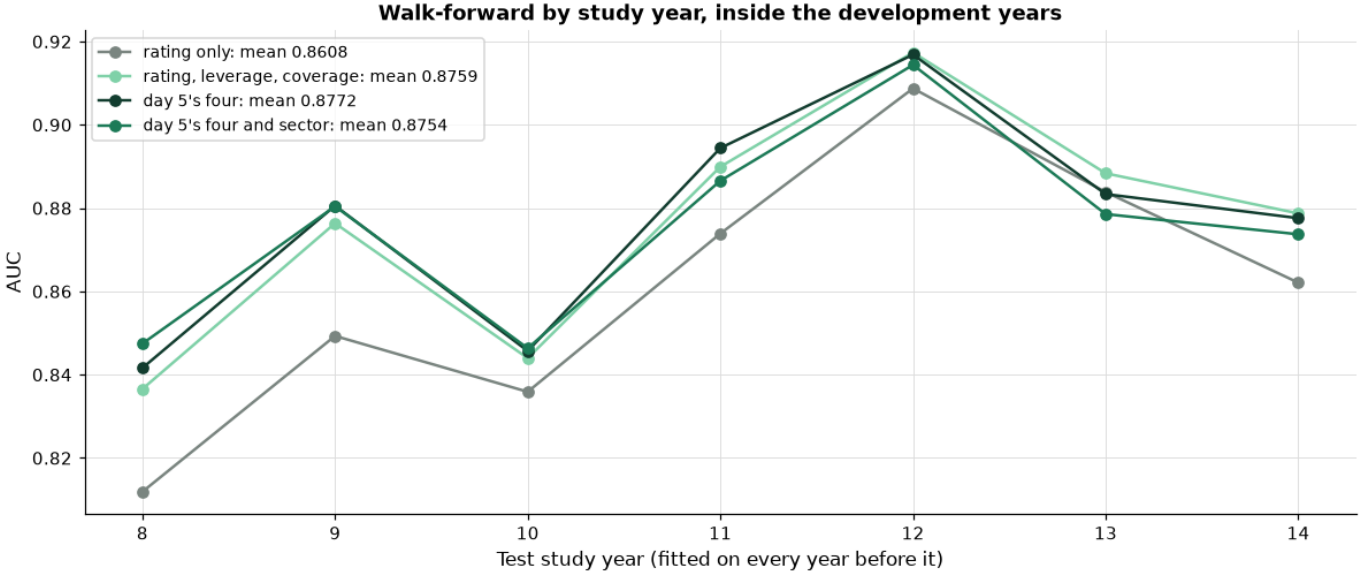
99% one-day VaR US$4.5m, shortfall US$5.8m. Rolling 500-day VaR: 28 exceptions in 1,500 days against 15 for a correct VaR; Kupiec p-value 0.0026.

Recorded in the choice file as it stands, not tuned away.

How the choice rule works

The input	A table: one row per fold, one column per model (higher is better), and the baseline's name
The rule	A model replaces the baseline only if its mean is higher AND it is higher in at least min_folds folds
The winner	Among those that qualify, the highest mean; with none, the baseline stays
The refusals	A baseline that is not a column, min_folds out of range, a blank
What comes out	A column name. A held-out score never goes in

The hazard model, chosen on development

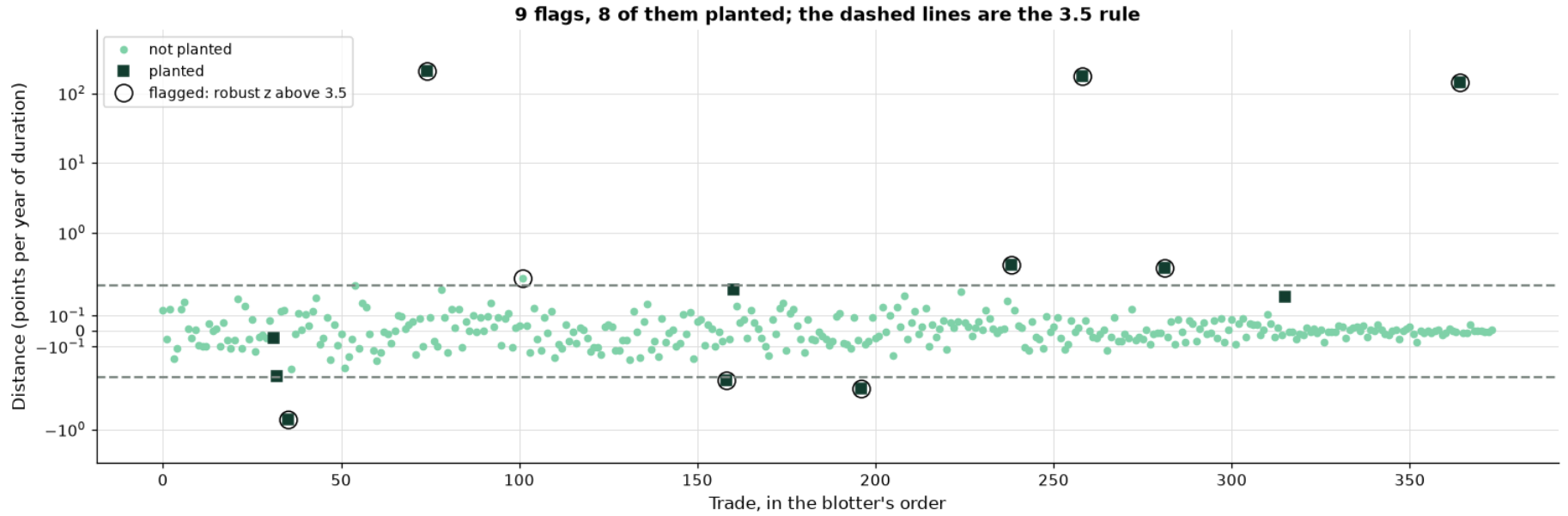


Model	Mean AUC	Above
rating only	0.8608	0
rating, leverage, coverage	0.8759	7
day 5's four	0.8772	6
day 5's four and sector	0.8754	6

Baseline: rating only. A majority: 4 of 7 years. Chosen: day 5's four, fitted once on study years 1 to 14 and frozen.

Development study years only. The held-out years are still closed.

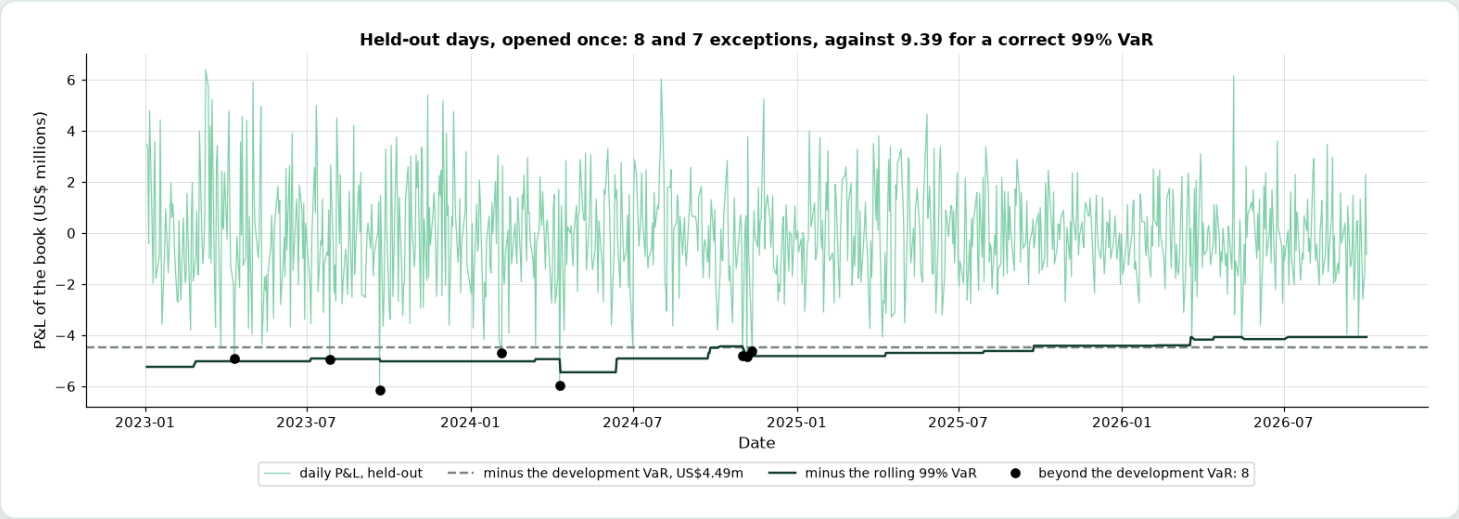
The anomaly screen



Rule	Flags	Planted	Precision	Recall
robust z above 3.5	9	8	0.889	0.667
outer fences (k = 3)	6	6	1.000	0.500

12 prints planted in a copy: 9 off-market, 3 typed ten times too large. The labels score the rules; they choose nothing.

The held-out years: the VaR

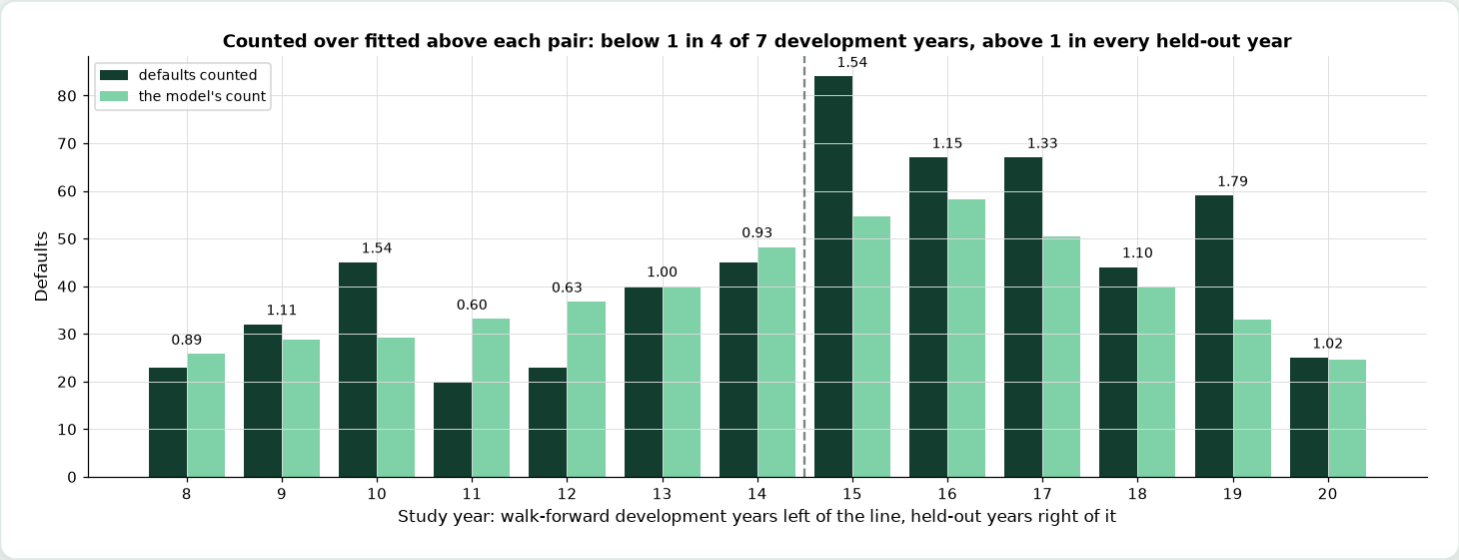


VaR	Count	Correct	p-value
95%, fixed	58	46.95	0.1099
95%, rolling	26	46.95	0.0006
99%, fixed	8	9.39	0.6399
99%, rolling	7	9.39	0.4117

939 held-out days, 3 Jan 2023 to 2 Oct 2026, opened once. At 99% neither VaR was rejected; at 95% the rolling VaR had too few exceptions.

The rolling VaR rejected inside development was not rejected on the held-out days.

The held-out years: the default levels

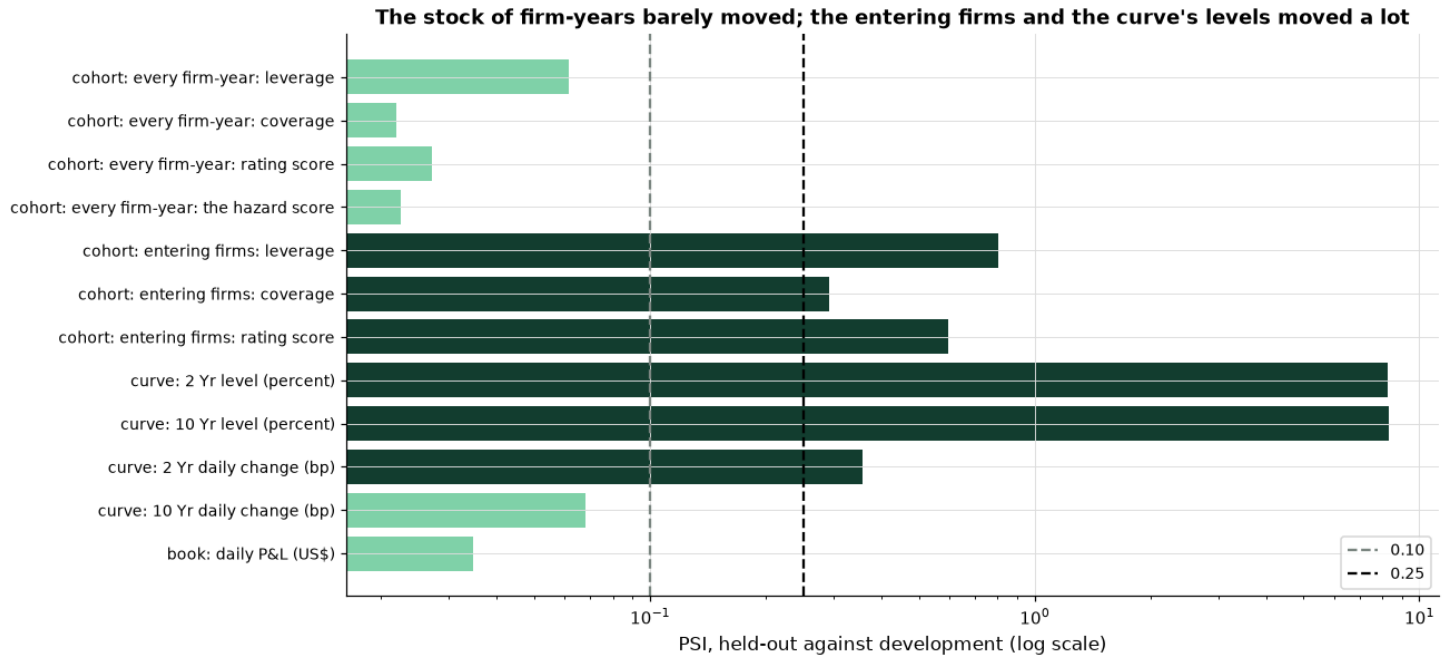


Bucket	Defaults	Rating model	Ratio
BBB	15	7.889	1.901
BB	45	37.599	1.197
B	167	113.924	1.466
CCC	119	92.625	1.285

Rating PDs: 346 defaults counted against 253.1, 1.367 times. Hazard model: AUC 0.8651, green; counted over fitted 1.3283, amber.

The ranking held; the level failed, in every held-out year.

What drifted



Stock and flow

Every firm-year: leverage 0.0613. Entering firms: leverage 0.8056, rating score 0.5962. 10 Yr level 8.3331, its daily change 0.0682; 2 Yr daily change 0.3565.

PSI describes how far two samples sit apart, never where the levels go next.

What the models show, and where they fail

Check	Development	Held-out
rolling 99% VaR: exceptions	28	7
exceptions a correct VaR averages	15	9.3900
Kupiec p-value	0.0026	0.4117
hazard model: AUC	0.8772	0.8651
hazard model: counted over fitted	0.9428	1.3283
rating PDs: counted over fitted	0.9137	1.3671

What held

The VaR on the held-out days, and the hazard model's ranking of the firms.

What failed

Both default levels fitted on the cohort, in every held-out year. The entering firms moved most, unseen by the stock-wide checks.

One book held fixed; invented spreads, firms, and trades; not a view on any bond, rate, or firm.

Excel side by side

Excel side by side (tested in Excel)

`=-PERCENTILE.INC(B2:B2001,0.01)`	99% VaR US$4.5m
`=SUMPRODUCT(--(-C2:C940>F1))`	8 held-out exceptions
`=CHISQ.DIST.RT(K3,1)`	Kupiec p-value 0.6399
`=SUM(A2:A11113)/SUM(C2:C11113)`	counted over fitted 1.3283
`=1-POISSON.DIST(SUM(A2:A11113)-1,SUM(C2:C11113),TRUE)`	that many or more: 2.5e-7
`=SUMPRODUCT((H2:H11-F2:F11)*LN(H2:H11/F2:F11))`	PSI 0.8056

Every value equals Python's. Excel has no logistic regression, no one-factor simulation, and no historical_pnl.

Save the day: the order in the history

1. Module

riskkit and its tests, passing

2. Rule

choose_model, committed

3. Choices

choice.md, before any held-out row

4. Results

then the results and the pages

```
Deliverable 10: the AI-use log
Deliverable 9: what the models show, and where they fail
Deliverable 9: the risk report's numbers
Deliverable 7: the choices, made on the development years
Deliverable 6: the trades the screen flagged
Deliverable 5: choose_model and its test
Deliverable 1: the module, every test passing
```

The choices were committed before the results: Git's order says they were made first.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the VaR is taken over every day, held-out days included: doubling the test's invented held-out losses moves it from US$3 to US$4.

Code that runs is not the same as code that is right.

What you can do now

- Credit loss by Monte Carlo, with its error, VaR, and shortfall
- Historical VaR and its backtest, inside development and once held-out
- A hazard model chosen by a tested rule, frozen, read for ranking and level
- A trade screen scored against labels that chose nothing
- Choices committed; held-out years opened once; stock and flow apart

Course 6 Neural Networks